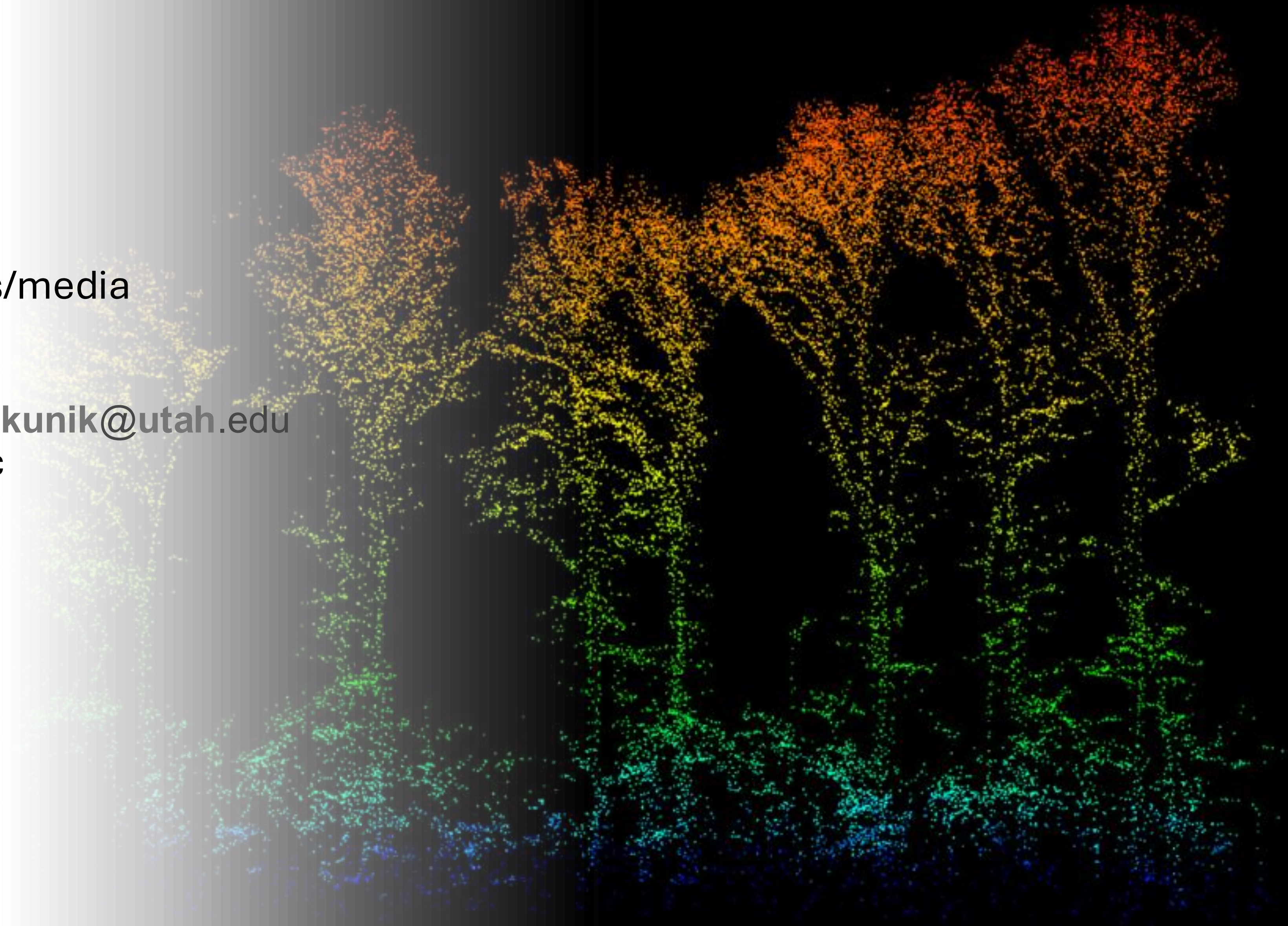
Week 1!

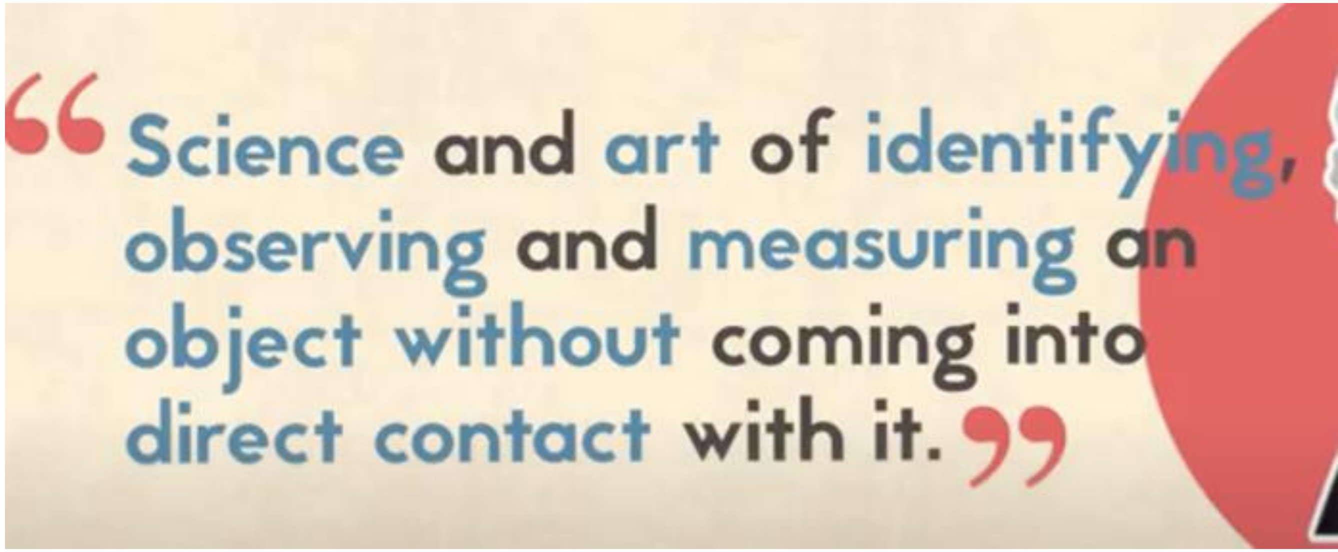
- Canvas layout (readings/media etc.)
- Quick note on ArcPro
- Intro to electromagnetic radiation
- Color is a lie
- Daily Q

lewis.kunik@utah.edu



First definition of remote sensing

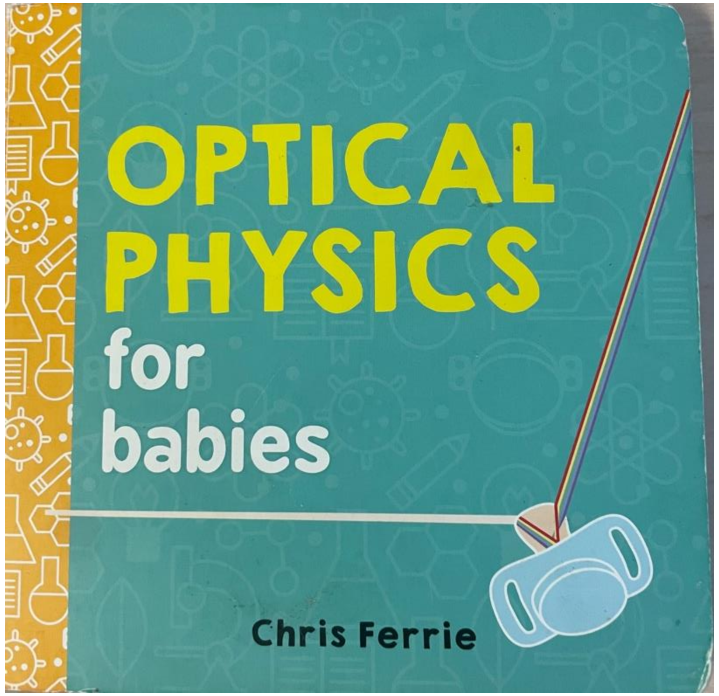
US Office of Naval Research

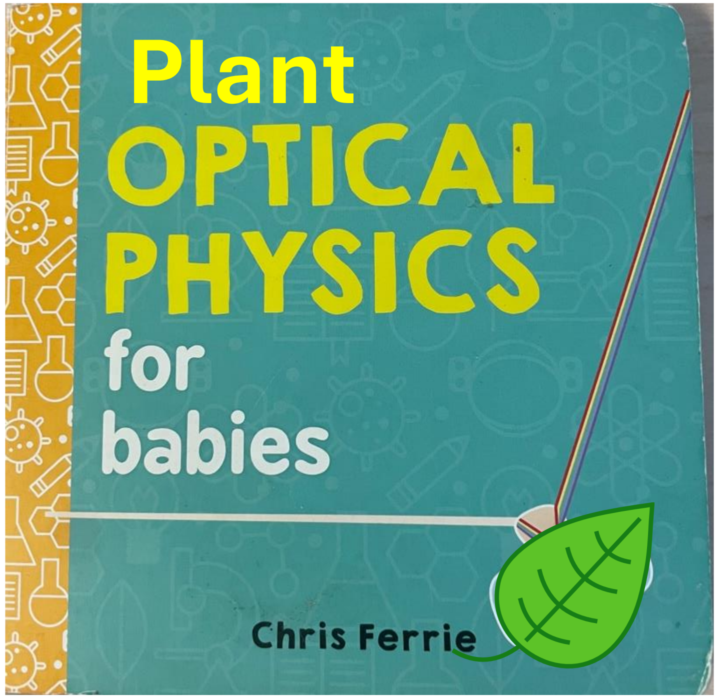


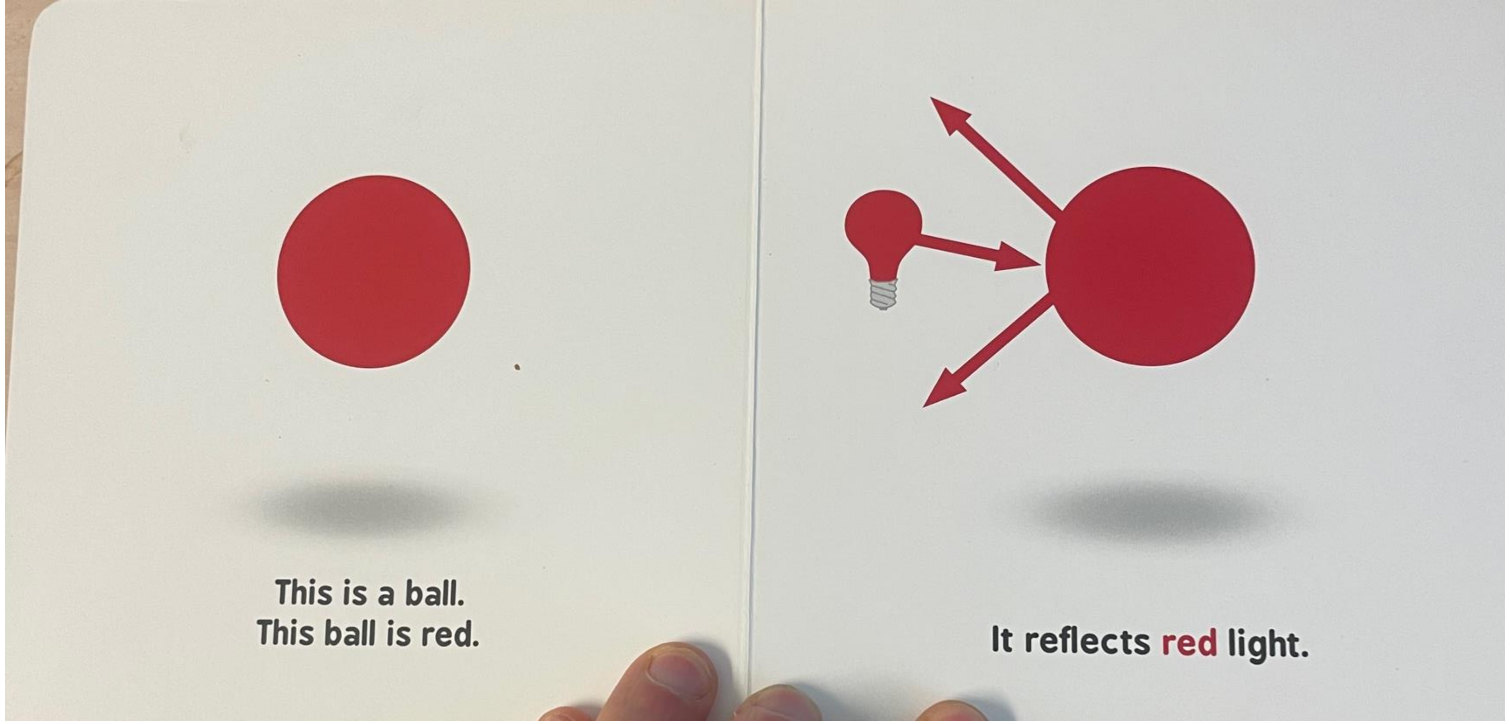
GIS is different because

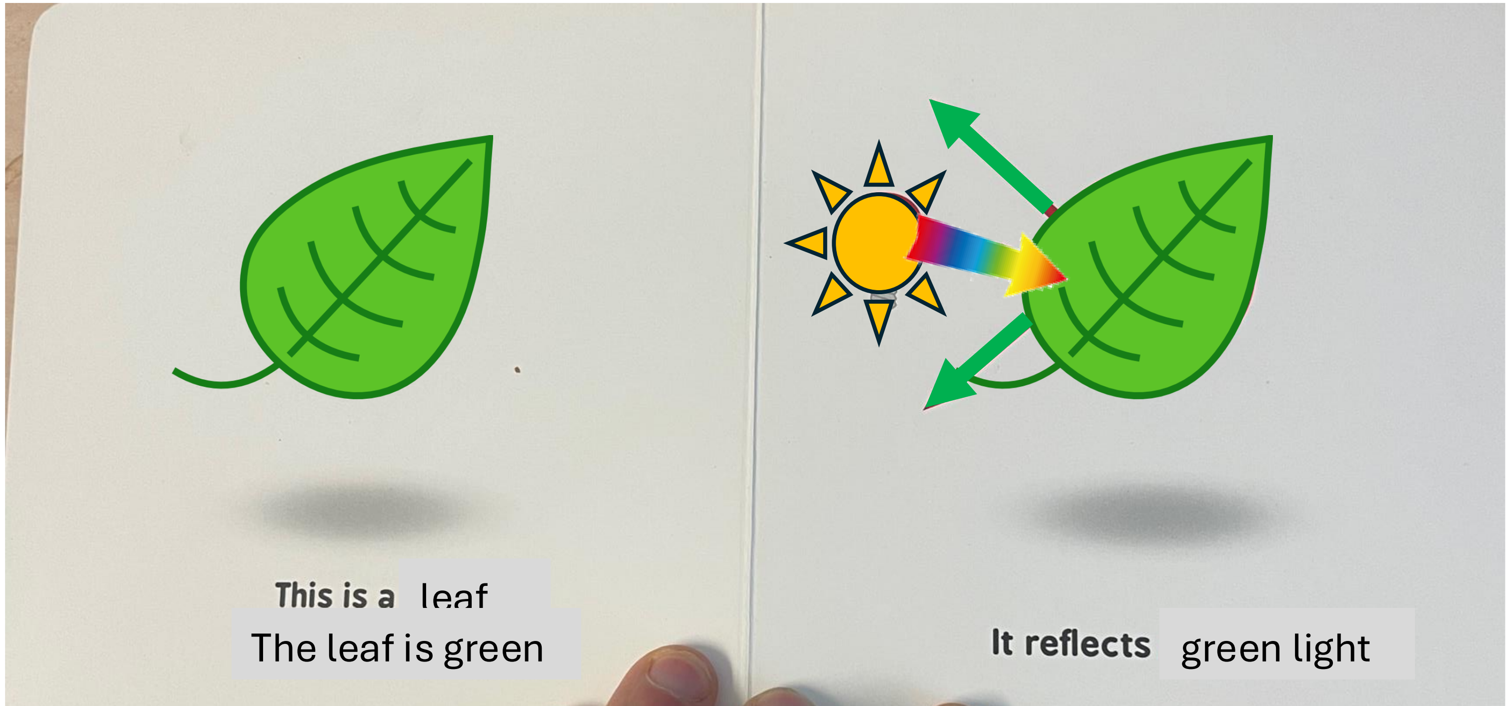
A GIS is a computerized system for:

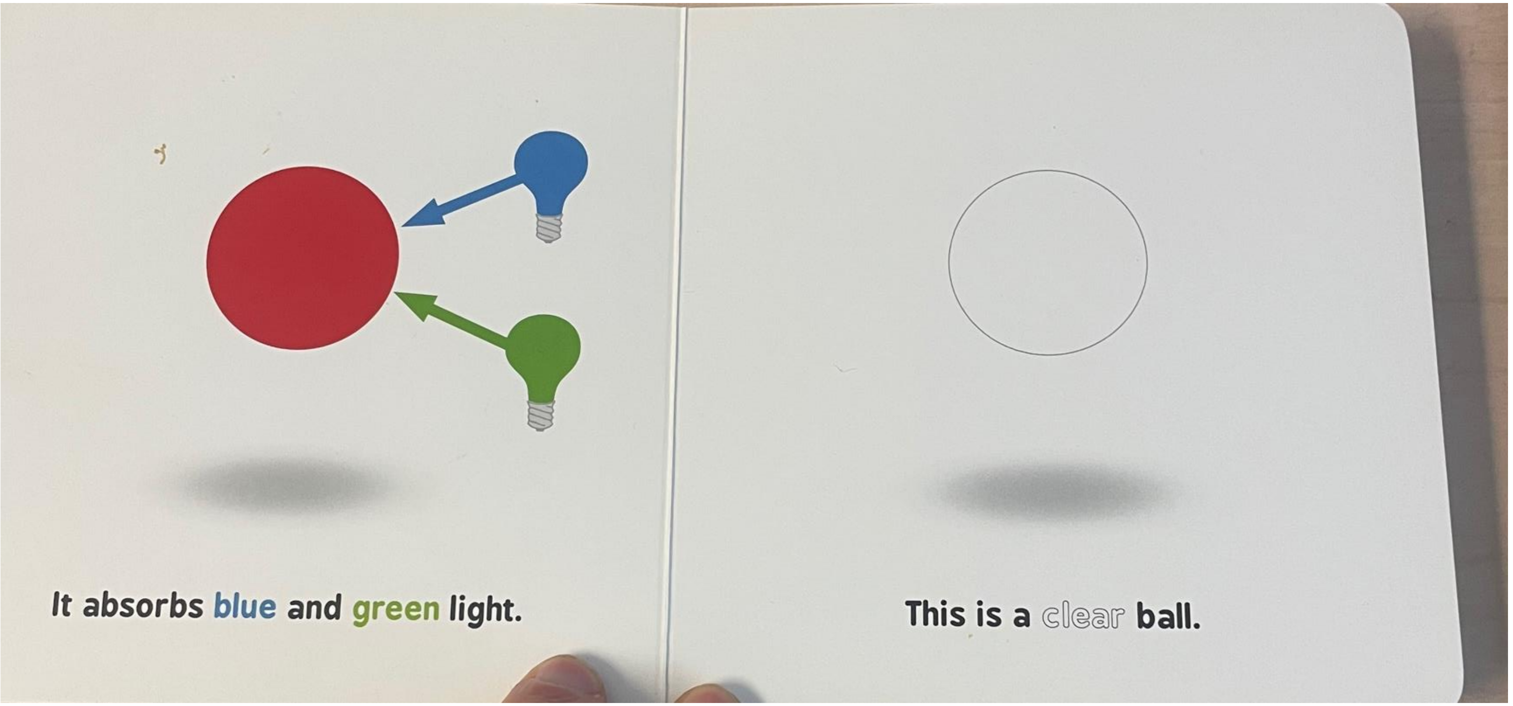
- compiling,
- storing,
- retrieving,
- analyzing, and
- displaying spatial data.

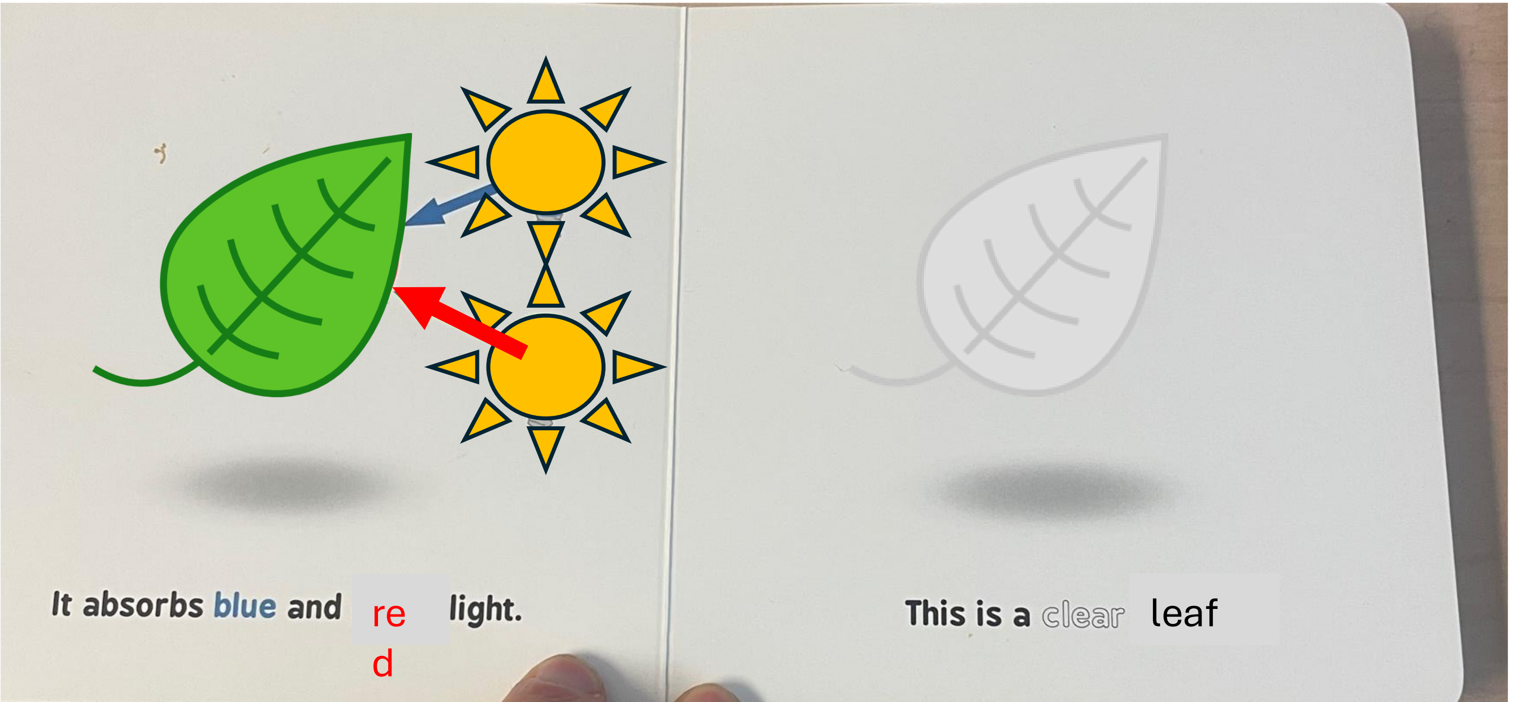

OPTICAL
PHYSICS
for
babies
Chris Ferrie

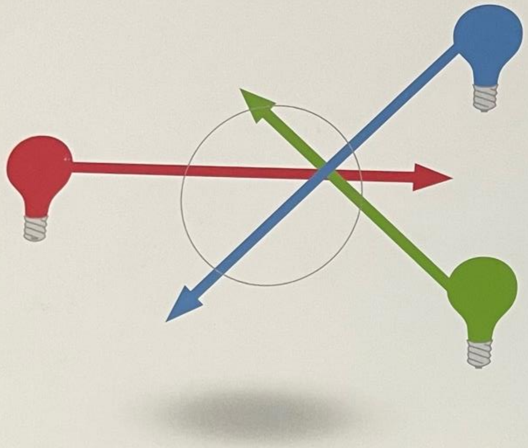

Plant
OPTICAL
PHYSICS
for
babies
Chris Ferrie


This is a ball.
This ball is red.
It reflects red light.

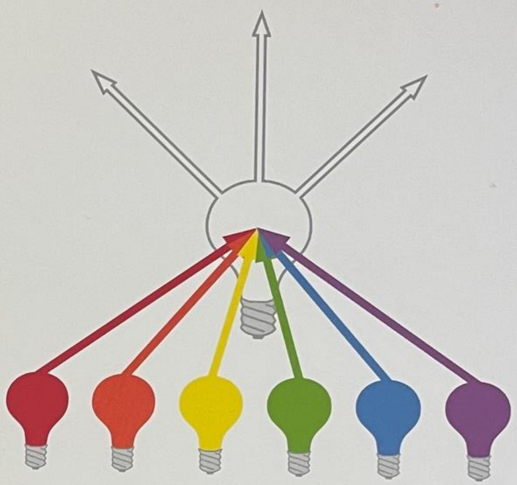

This is a leaf
The leaf is green
It reflects green light


It absorbs blue and green light.
This is a clear ball.

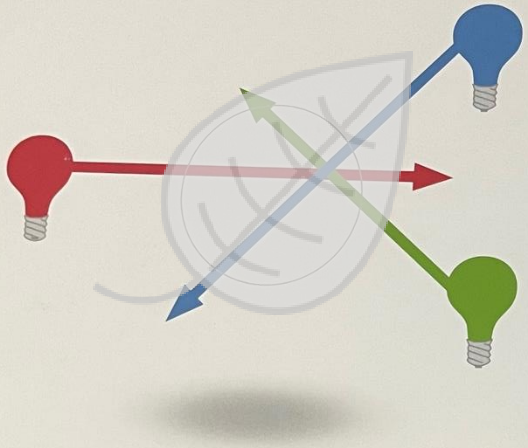

It absorbs blue and red light.
This is a clear leaf



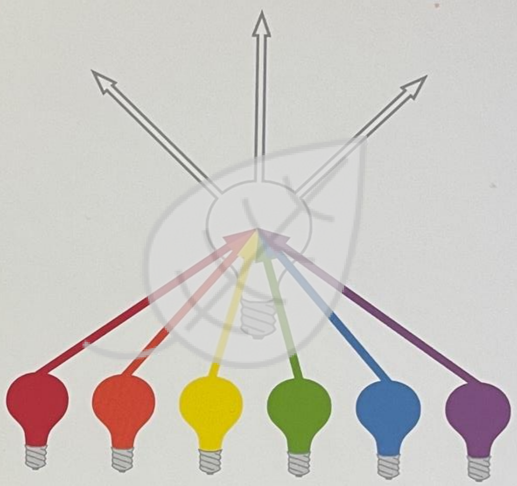
Colors pass through it.



All colors combine to make white light.



Colors pass through it.



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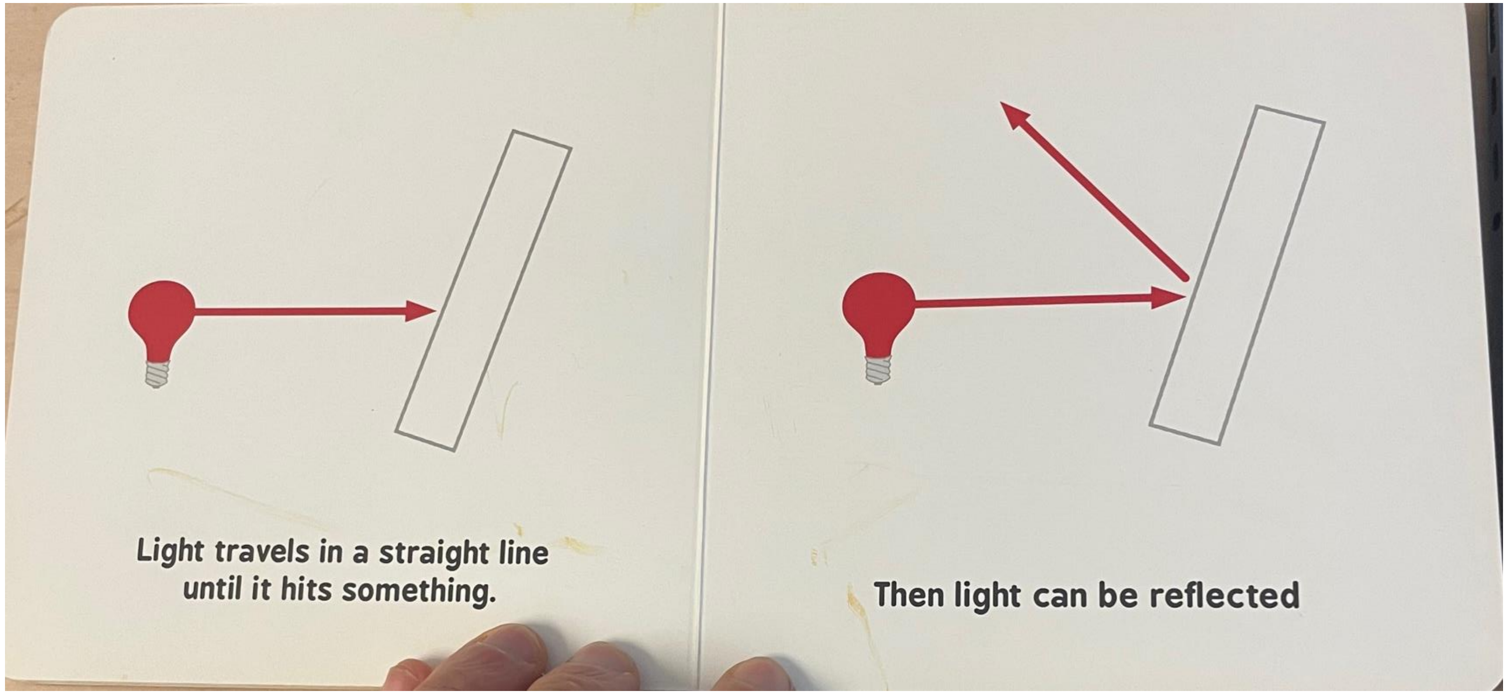
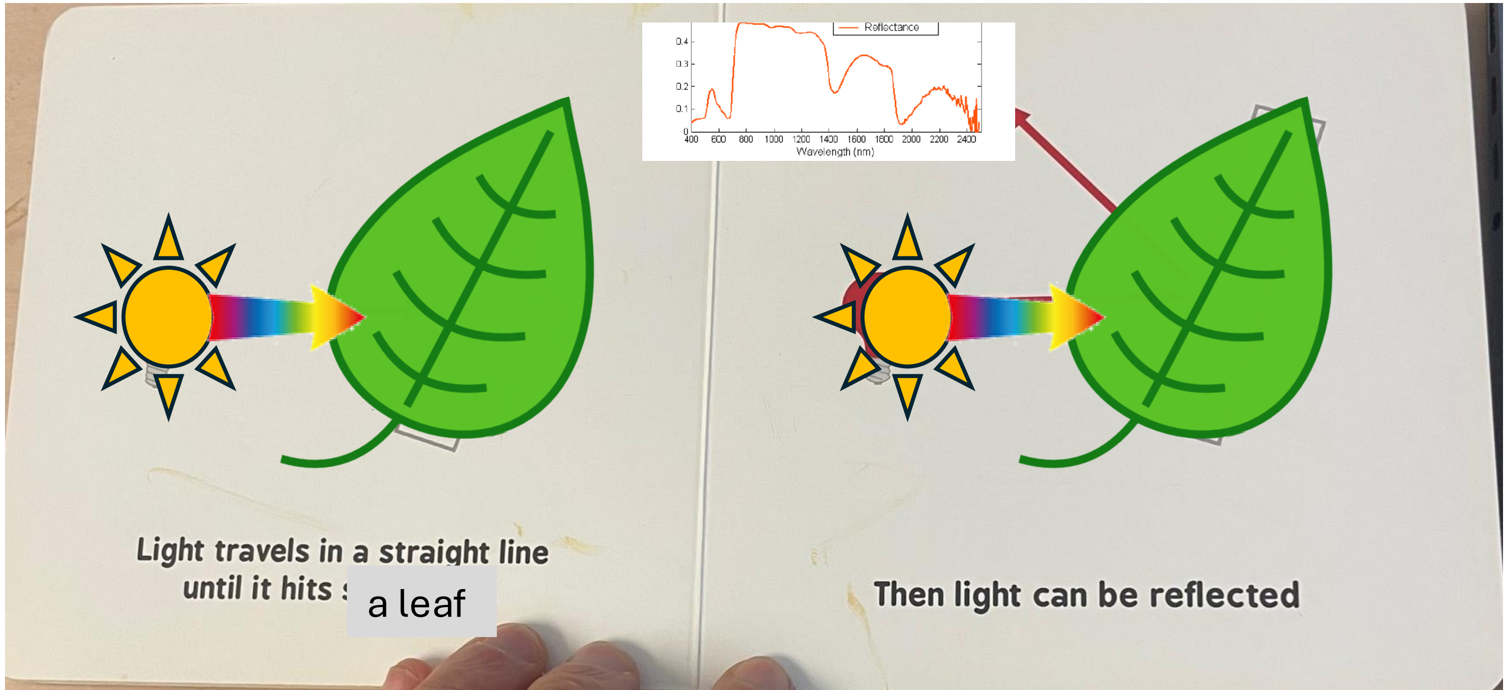
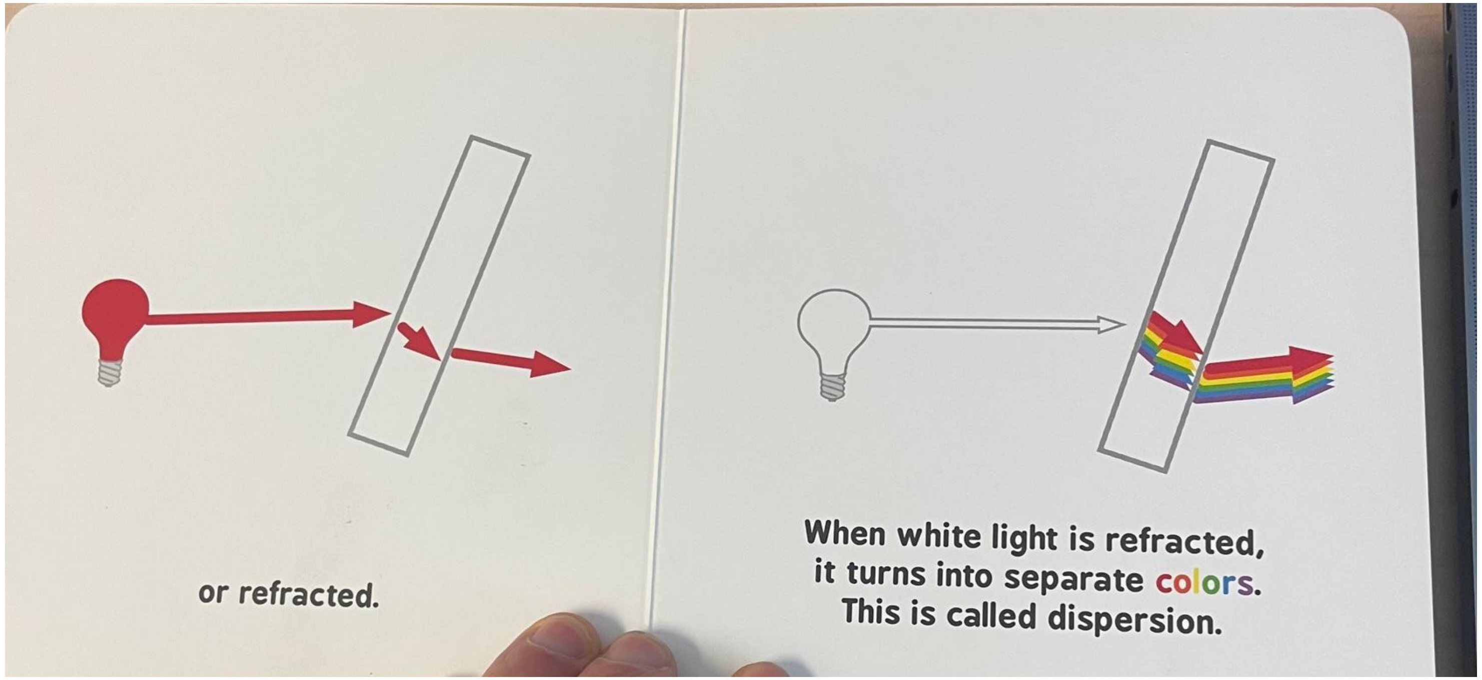
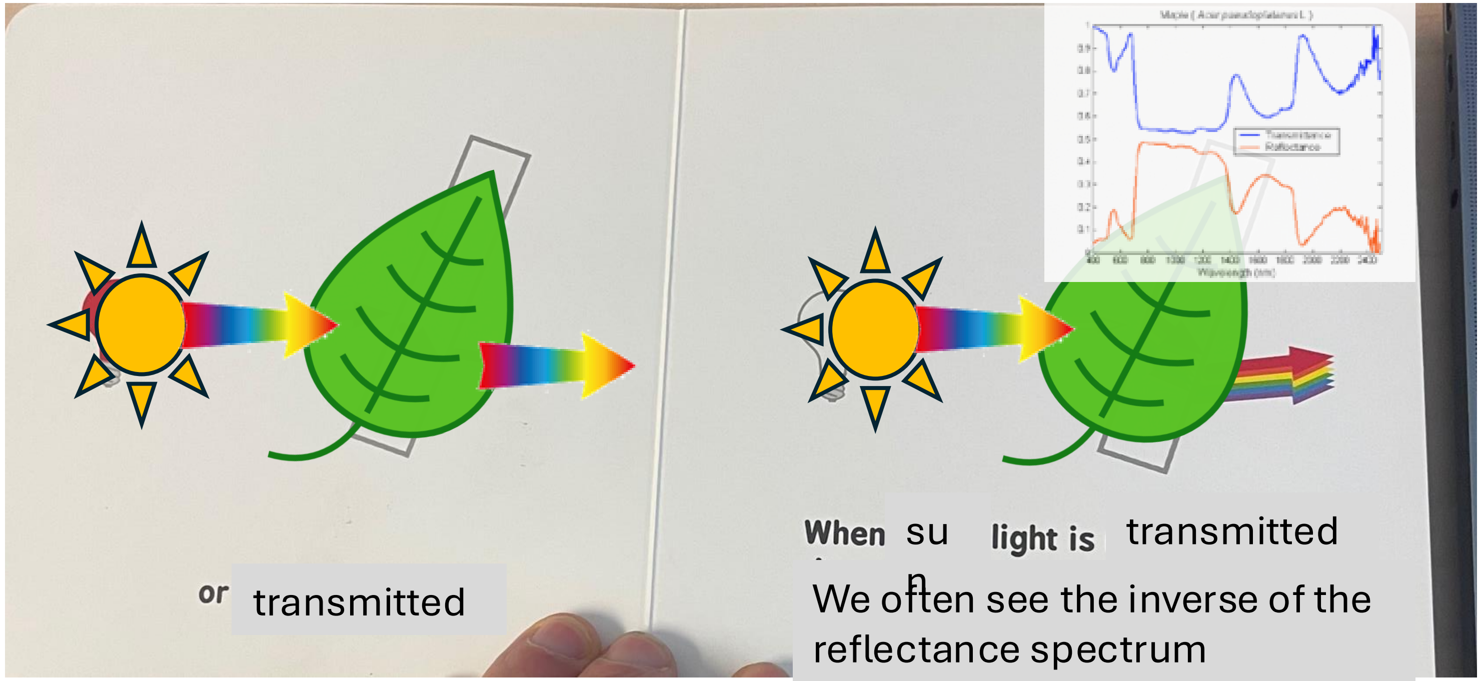
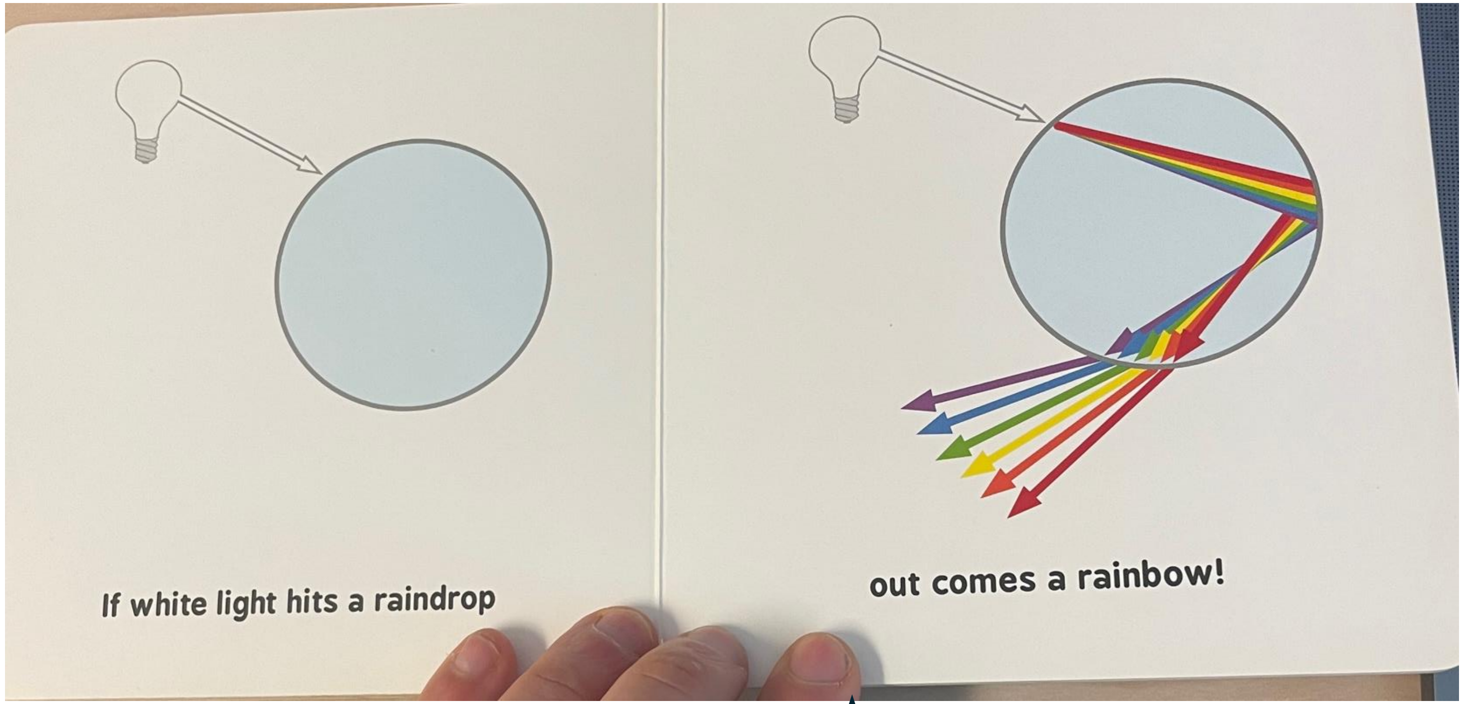
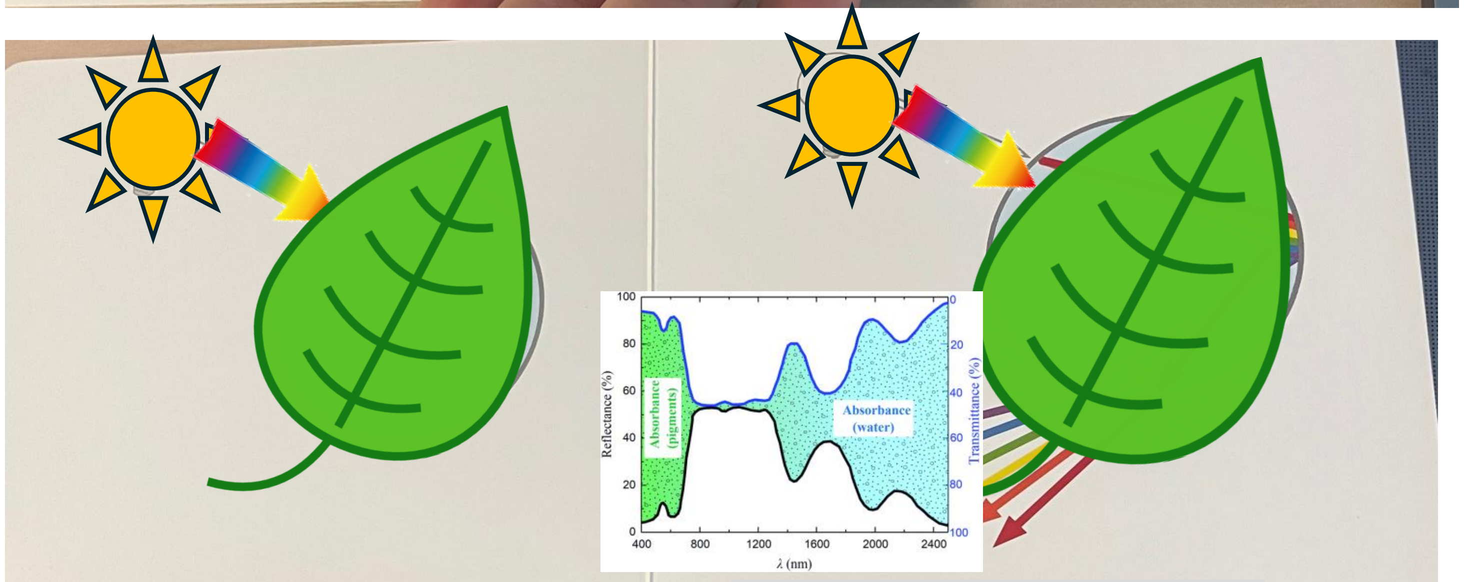

Light travels in a straight line
until it hits something.
Then light can be reflected


Reflectance
0.4
0.3
0.2
0.1
0
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
Wavelength (nm)
Light travels in a straight line
until it hits
a leaf
Then light can be reflected


or refracted.
When white light is refracted,
it turns into separate colors.
This is called dispersion.

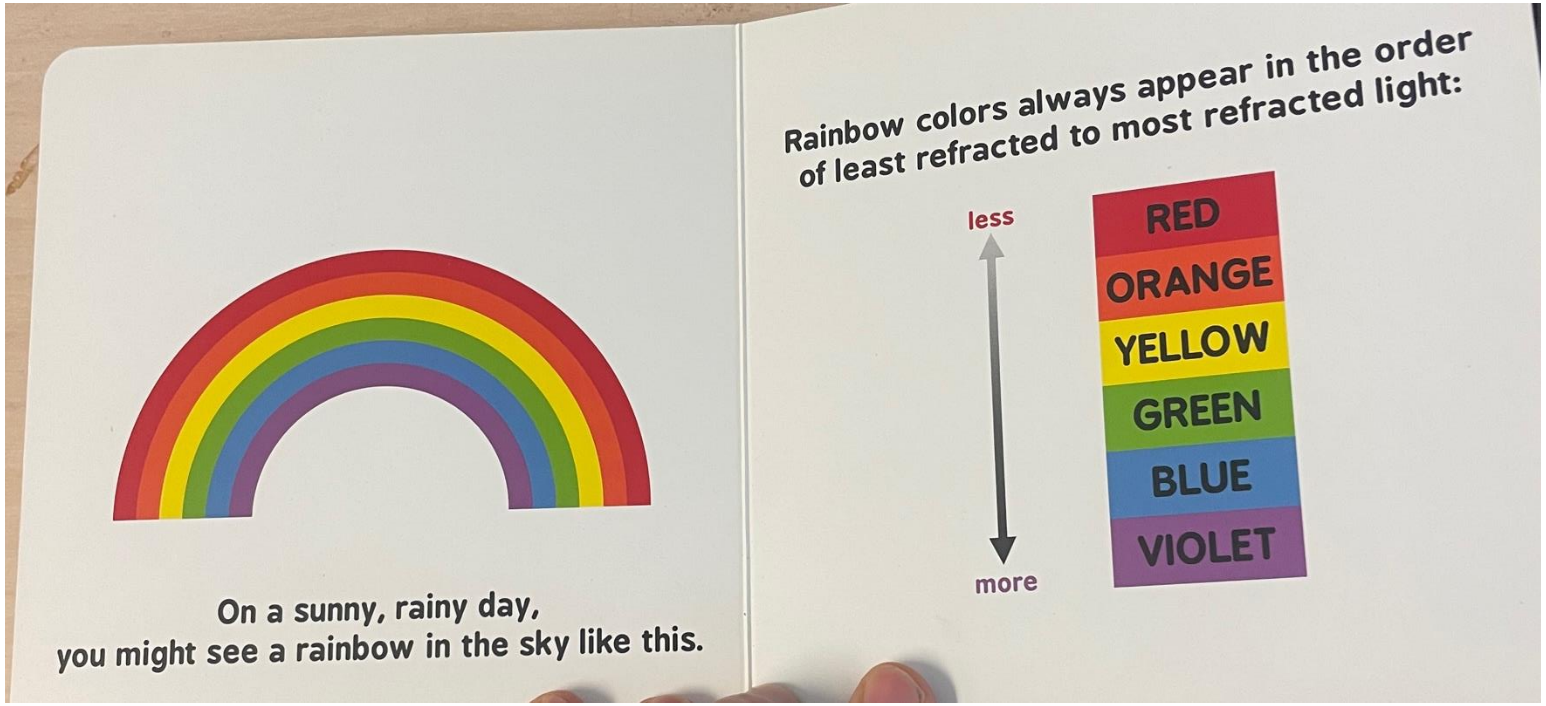

Maple (Acer pseudoplatanus L.)
Transmittance
Reflectance
Wavelength (nm)
or transmitted
When sun light is transmitted
We often see the inverse of the
reflectance spectrum



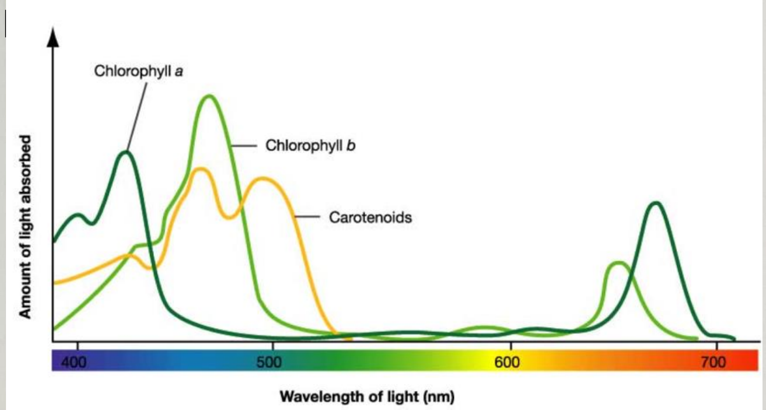


If a leaf has a lot of water in it

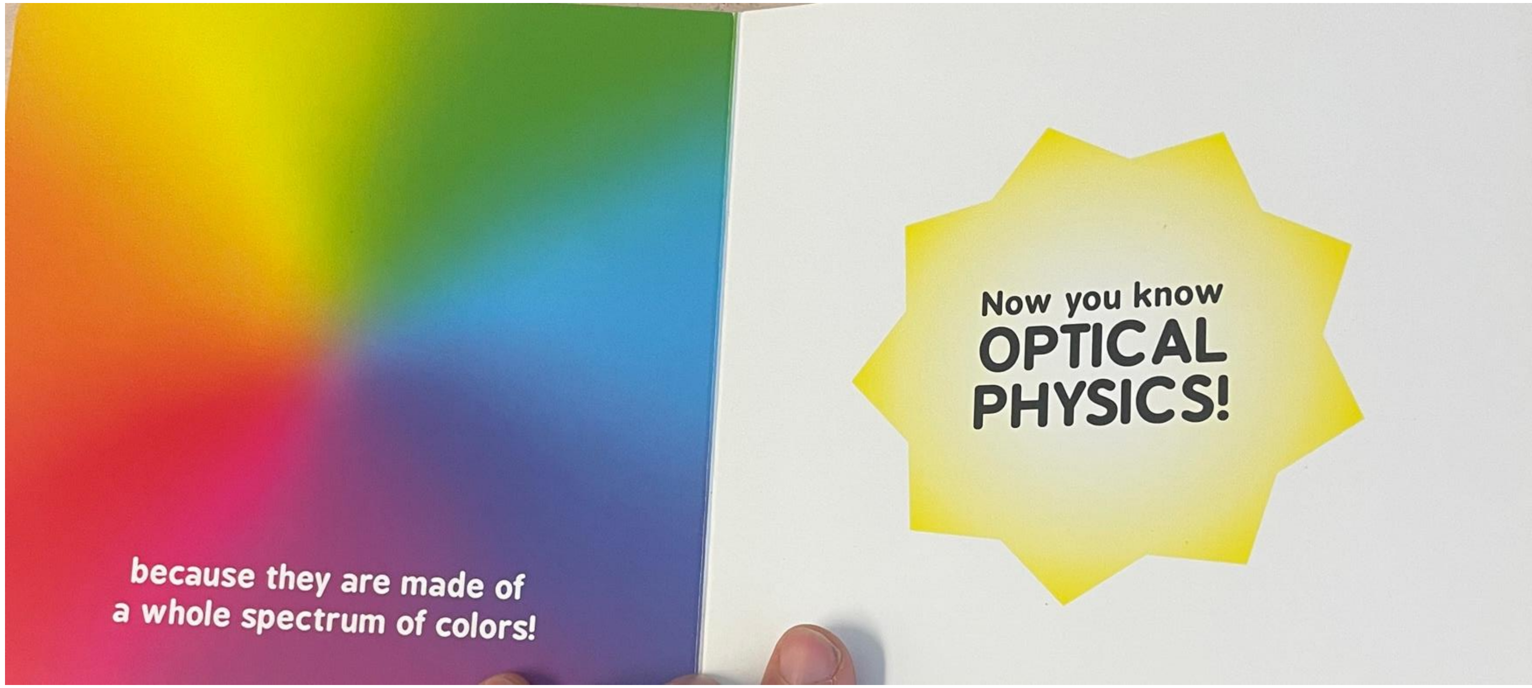
There’s more absorption in the shortwave infrared

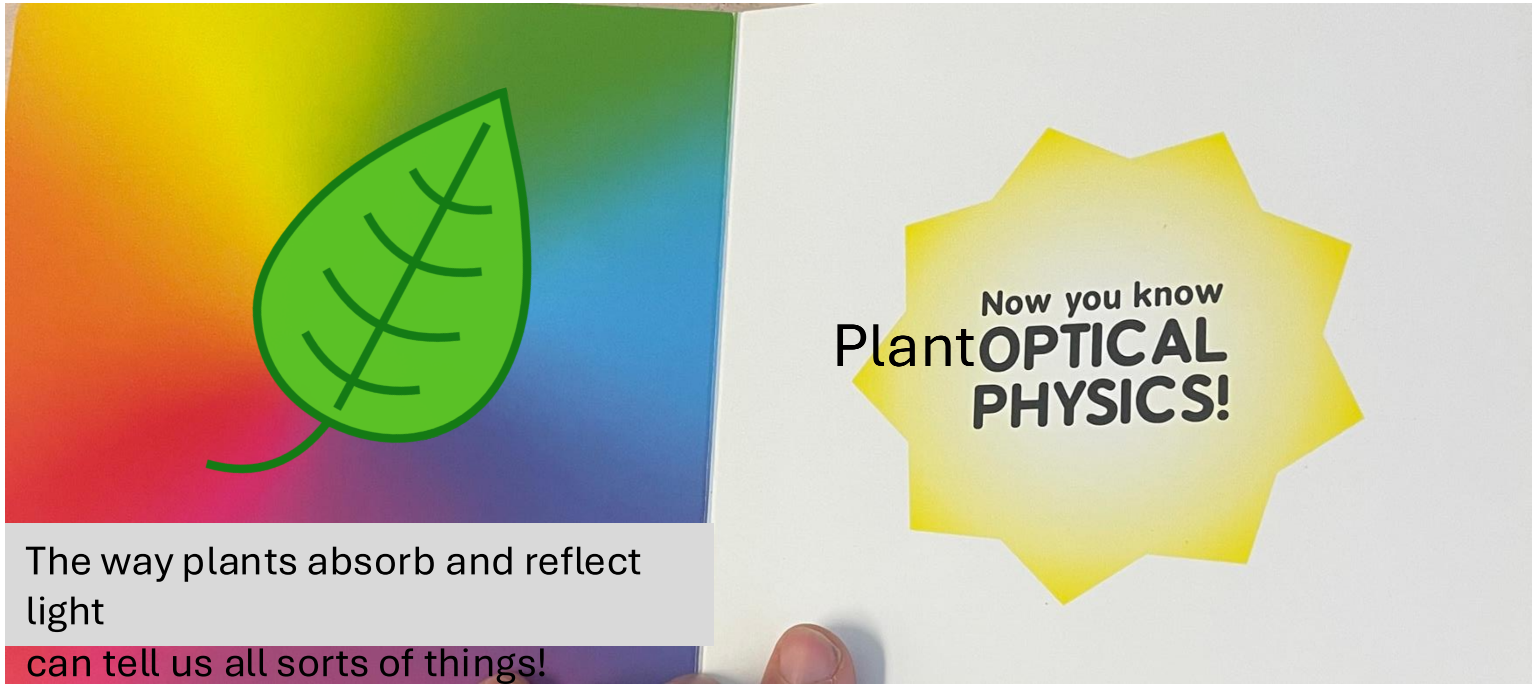


Plant pigments have absorption features that tell us how energy is being

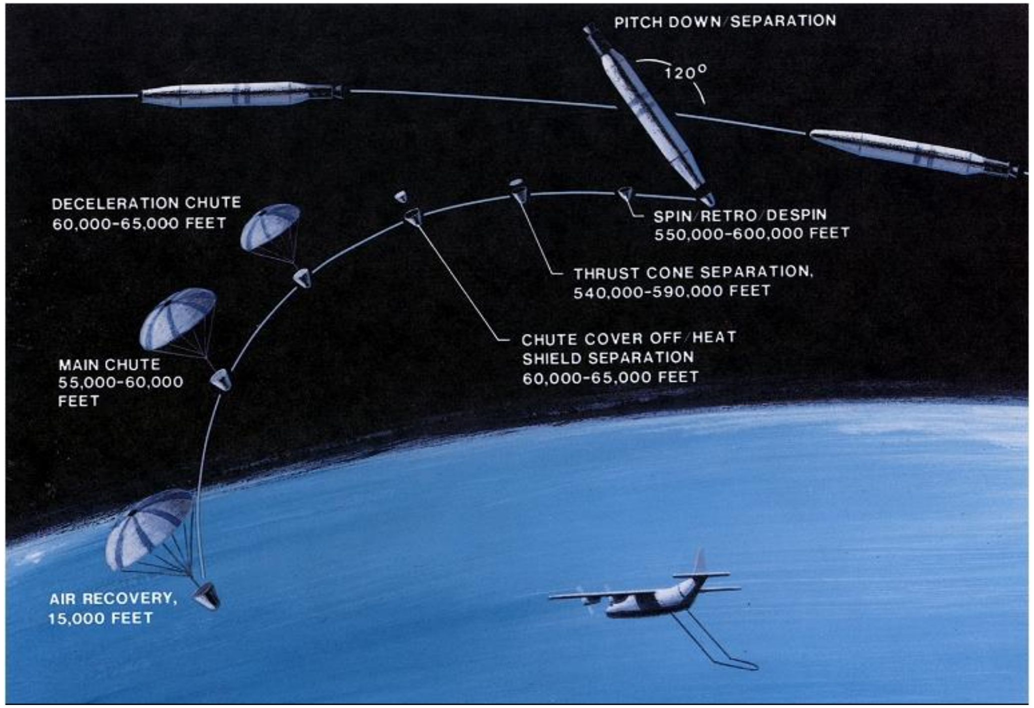


on any given day
plants will regulate light absorption for photosynthesis in the visible part of the spectrum


because they are made of
a whole spectrum of colors!
Now you know
OPTICAL
PHYSICS!


Now you know
PlantOPTICAL
PHYSICS!
The way plants absorb and reflect light
can tell us all sorts of things!

Seeing plant stuff from space



Corona satellites (1950s-1970s)

Estimation of USSR wheat yields in 1953

~~S-E-C-R-E-T~~

INTELLIGENCE MEMORANDUM

ESTIMATE OF 1953 GRAIN PRODUCTION
IN THE SOVIET BLOC

CIA/RR IM-395

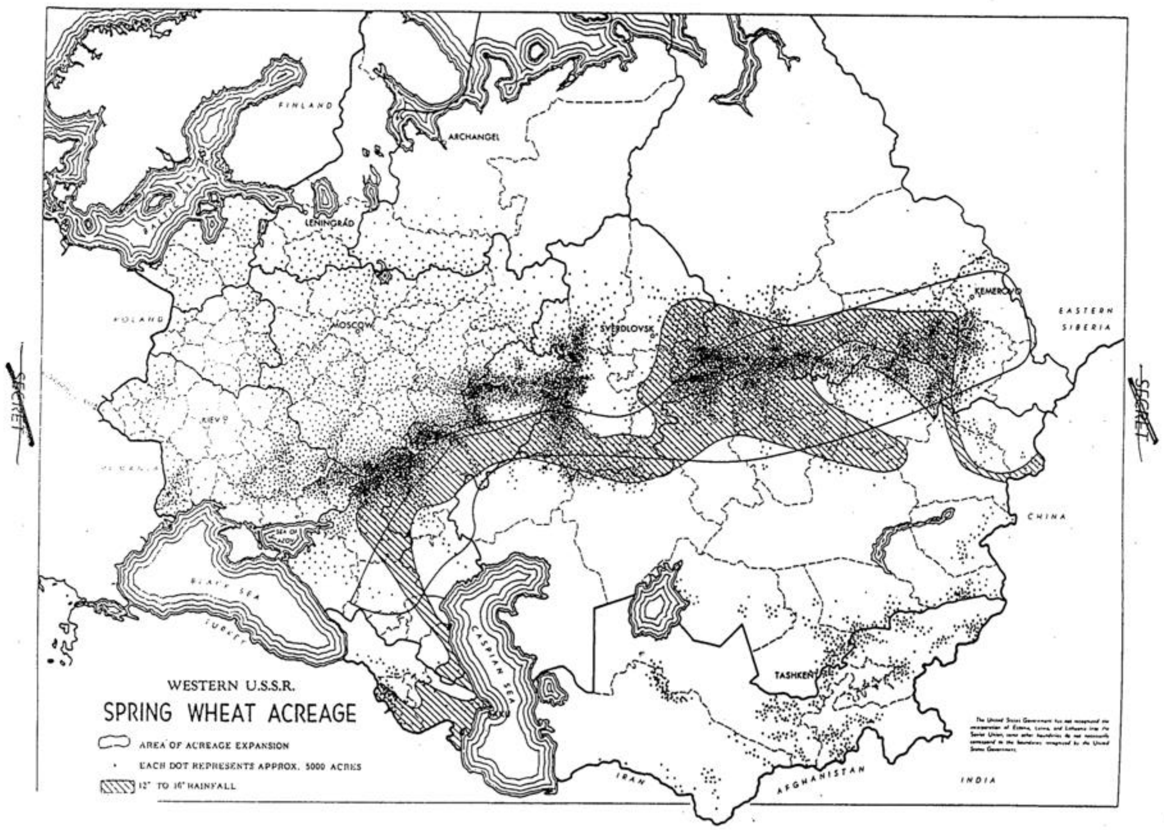
13 September 1954

WARNING

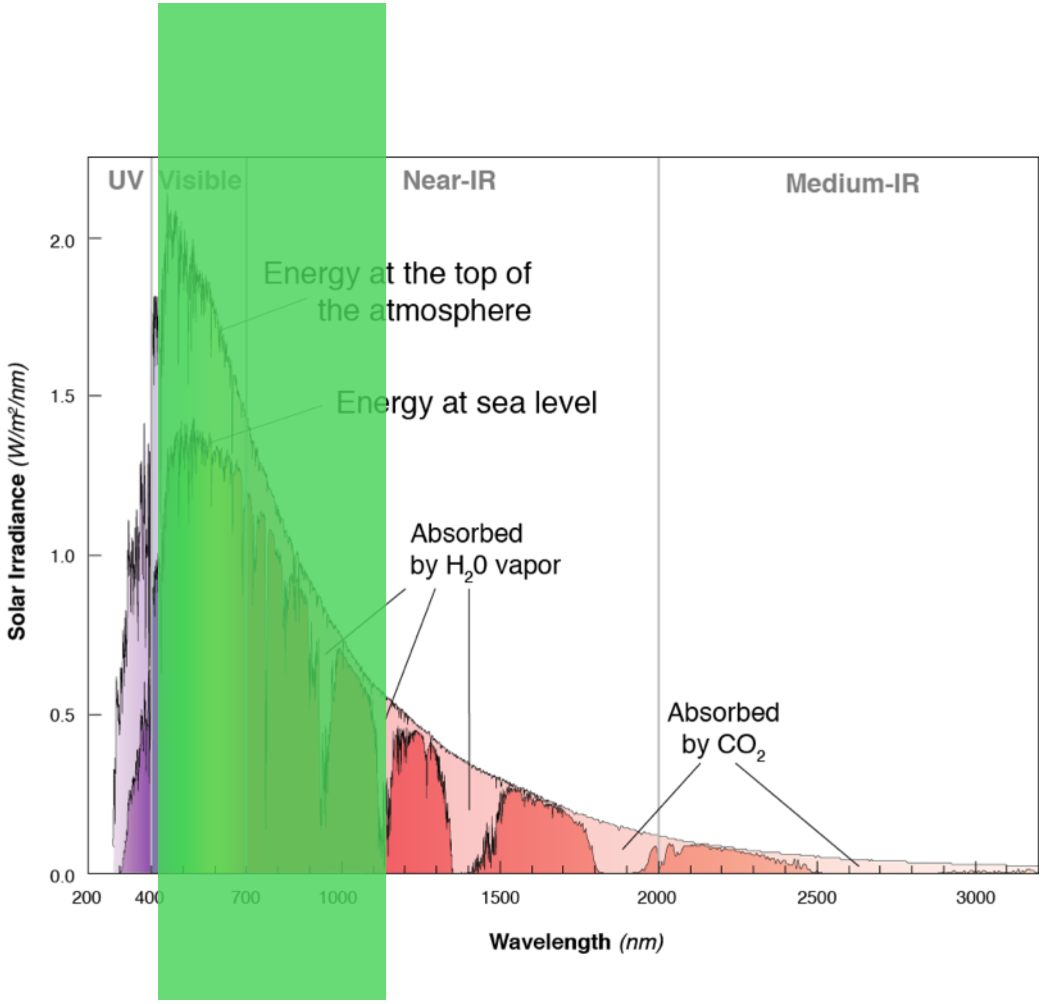
~~THIS MATERIAL CONTAINS INFORMATION AFFECTING THE NATIONAL DEFENSE OF THE UNITED STATES WITHIN THE MEANING OF THE ESPIONAGE LAWS, TITLE 18, USC, SECS. 793 AND 794, THE TRANSMISSION OR REVELATION OF WHICH IN ANY MANNER TO AN UNAUTHORIZED PERSON IS PROHIBITED BY LAW.~~

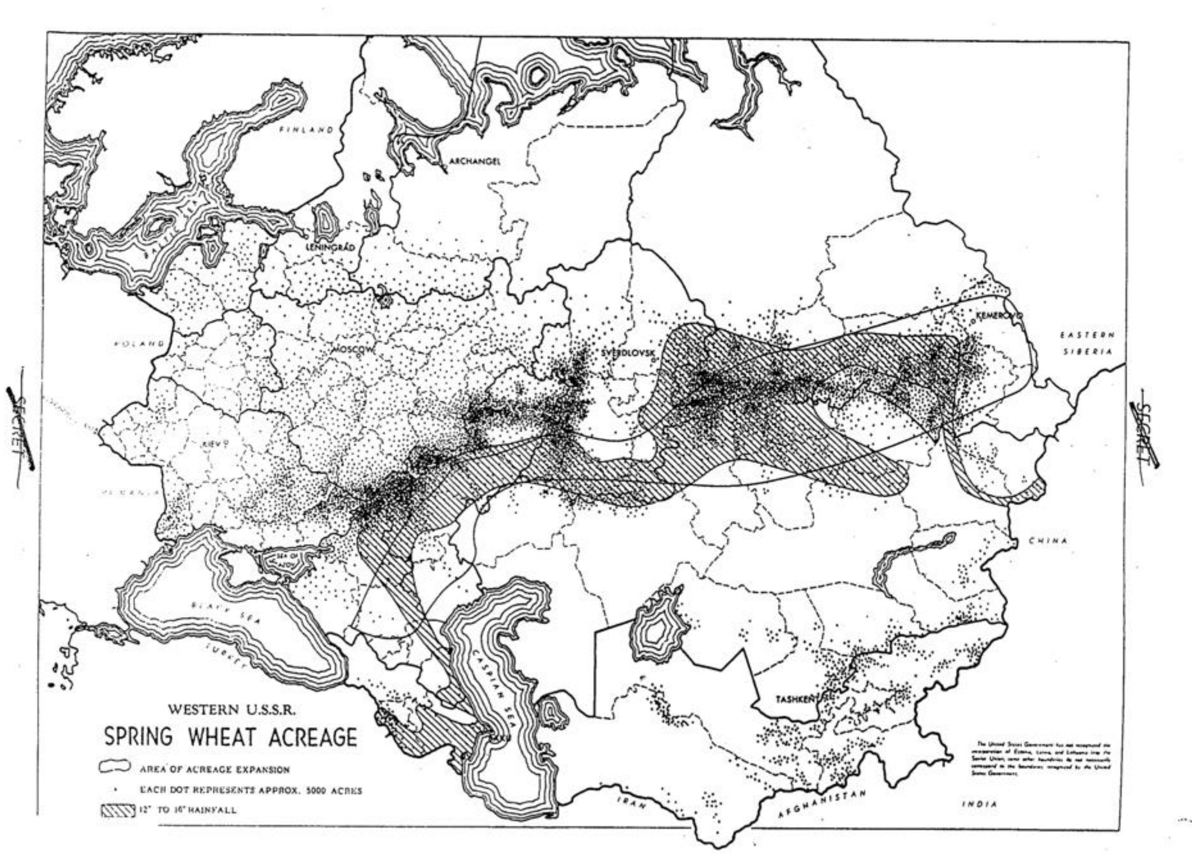
CENTRAL INTELLIGENCE AGENCY

Office of Research and Reports



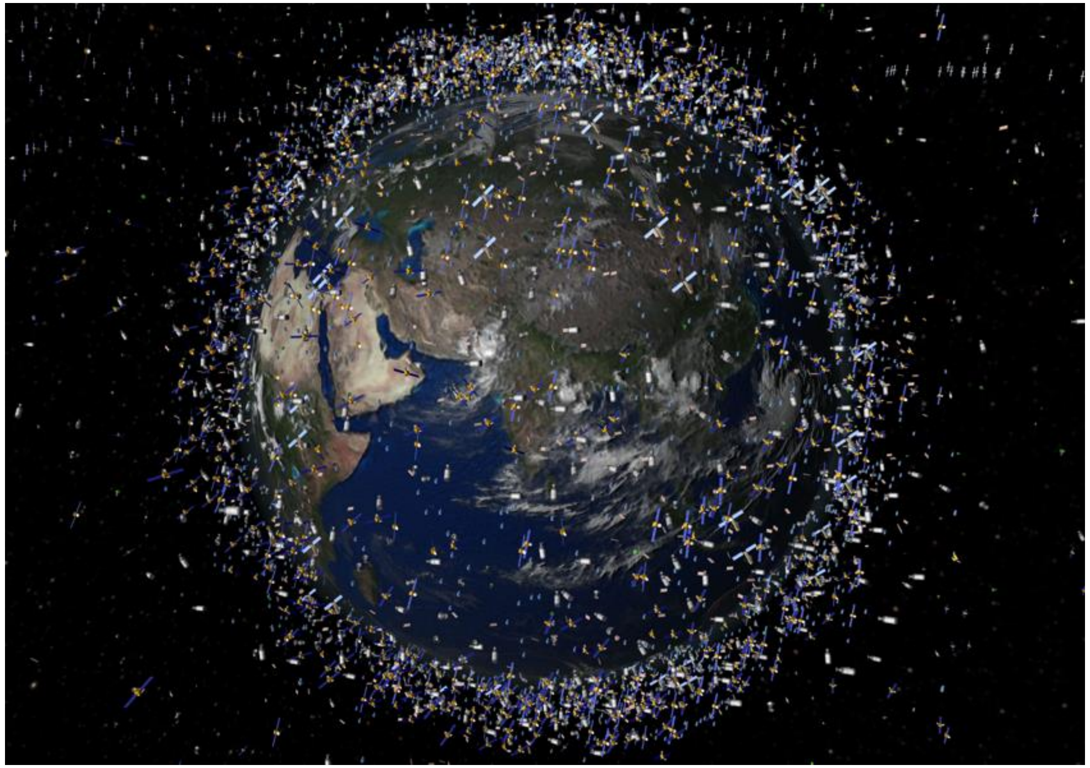
Estimation of USSR wheat yields in 1953



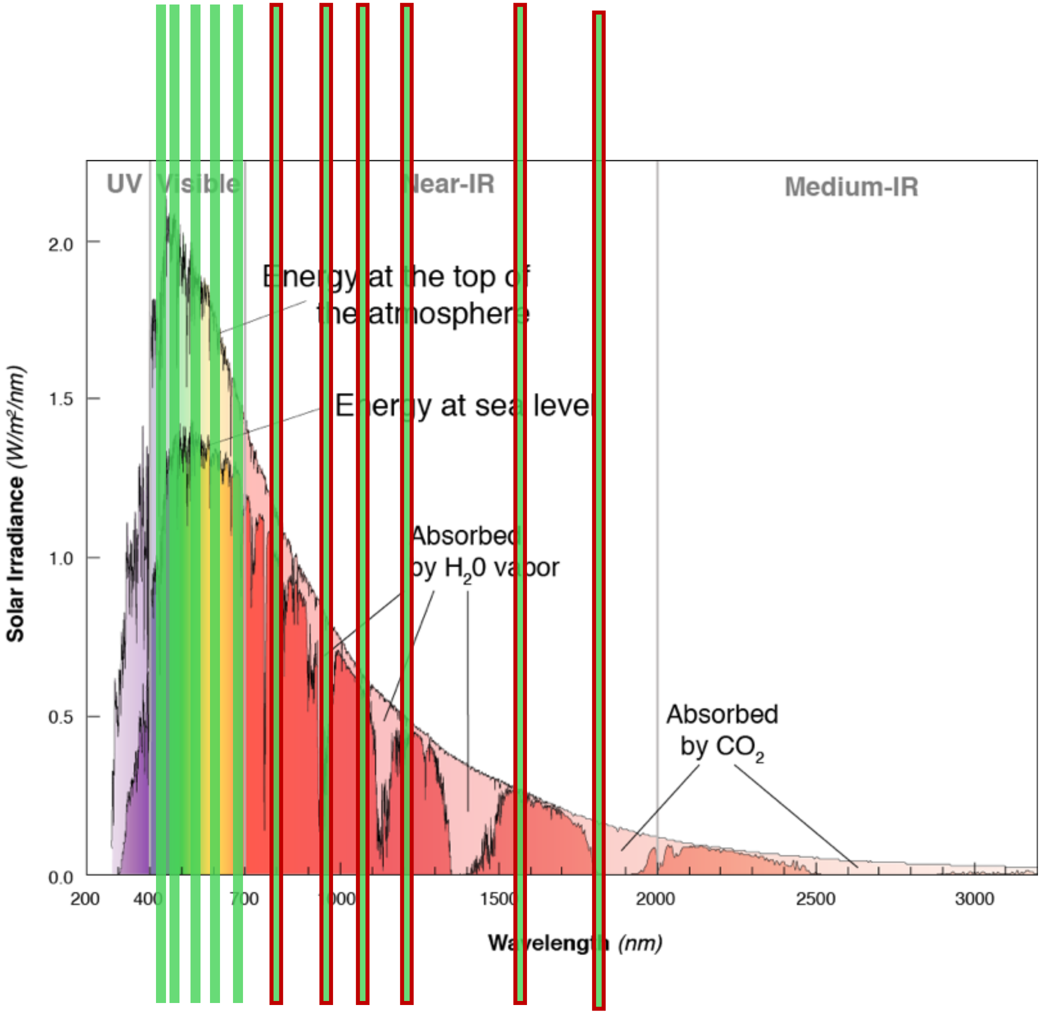


Measuring the ‘broad’ spectrum…film

Fast forward to today

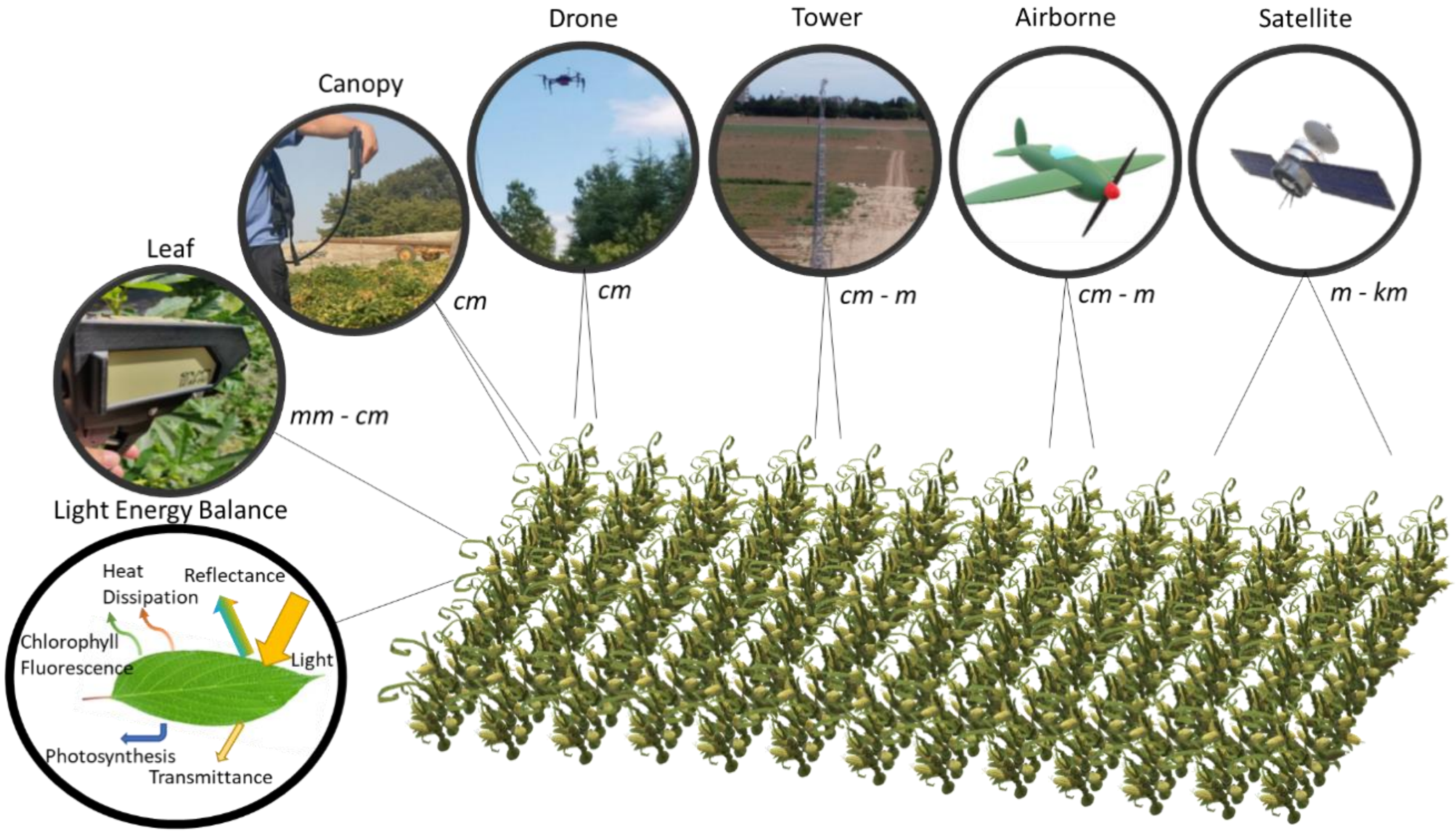


Nearly ~10,000 satellites orbiting earth



Measuring ALL over the place

Remote sensing across scales



Credit: Chris Wong

Active vs. Passive sensors

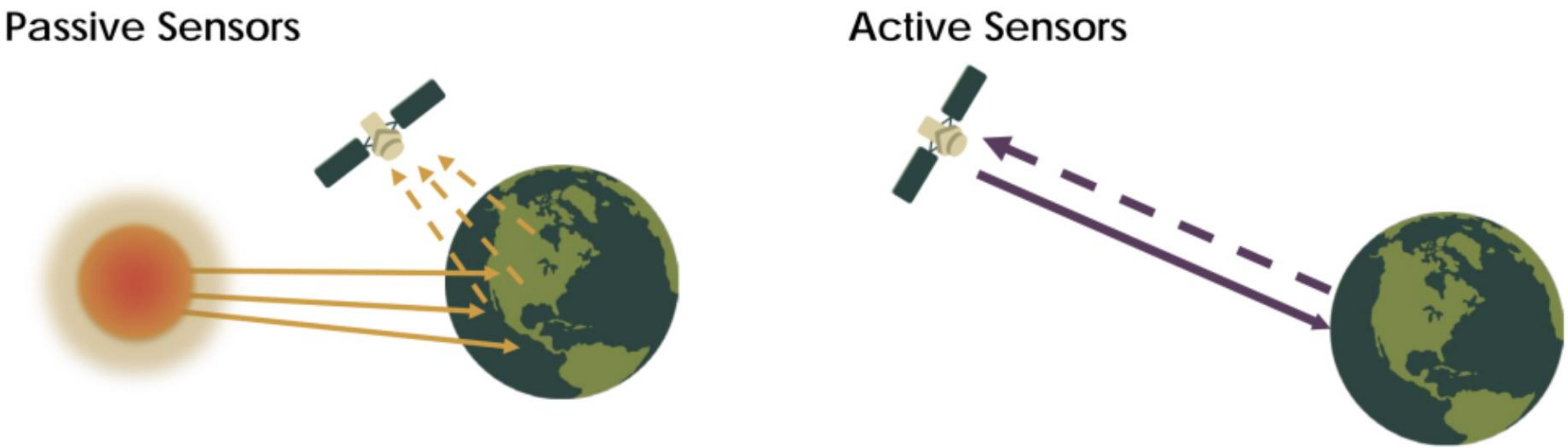
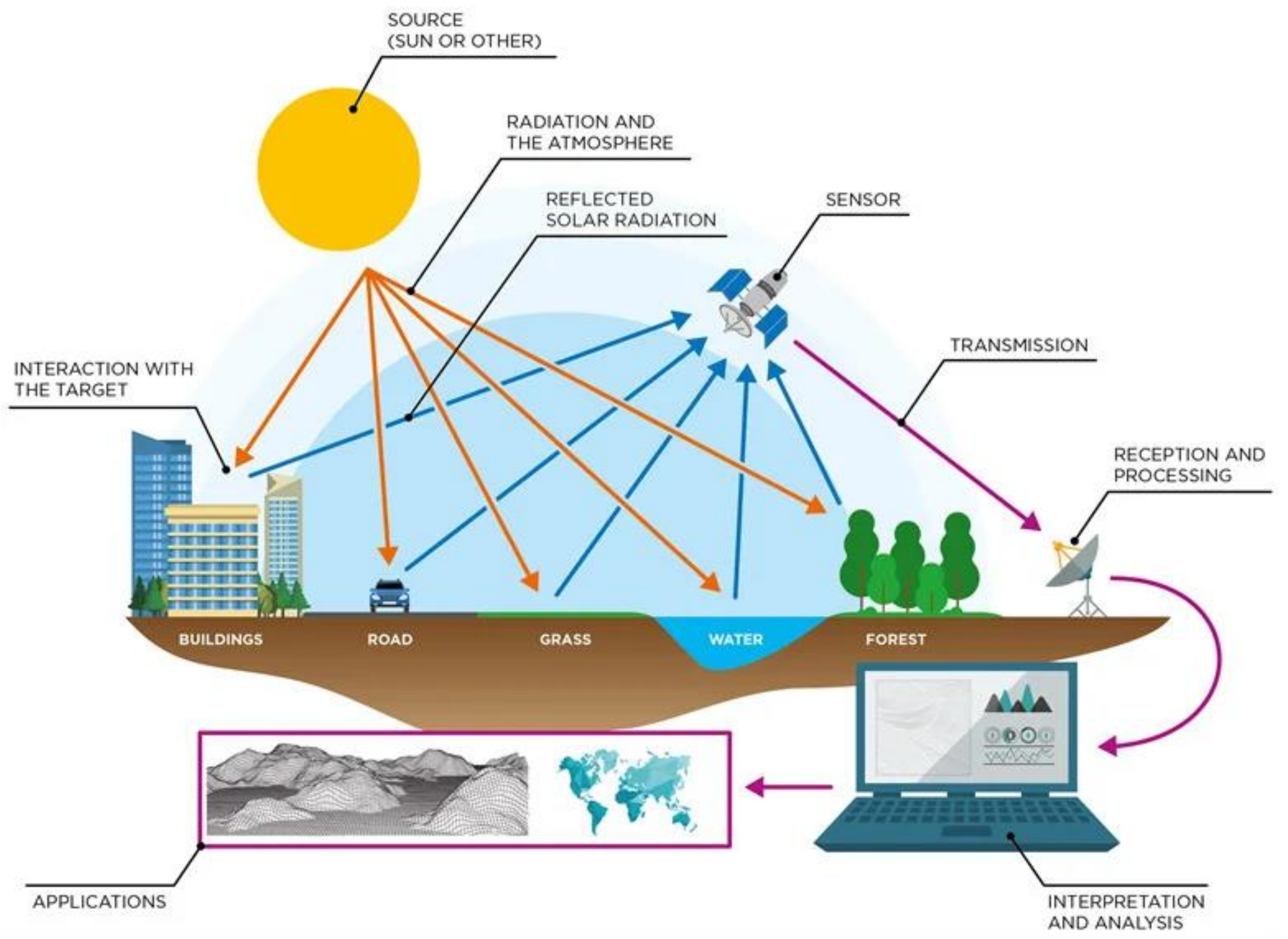
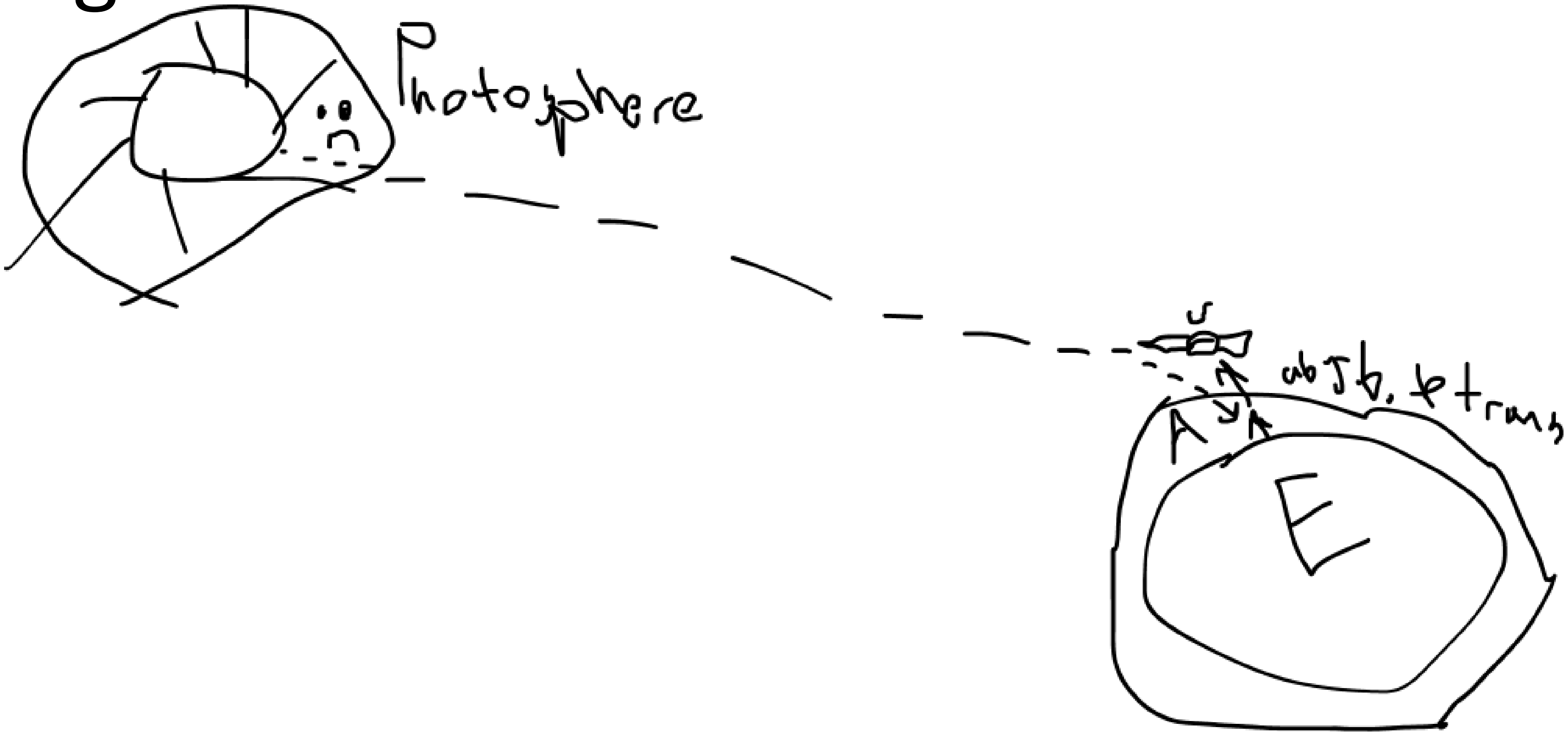


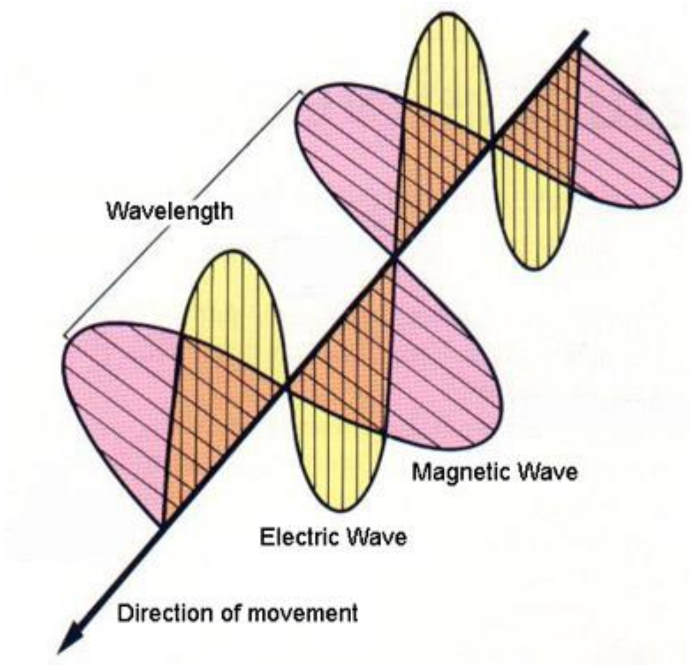
Diagram of a passive sensor versus an active sensor. Credit: NASA Applied Sciences Remote Sensing Training Program.


SOURCE
(SUN OR OTHER)
RADIATION AND
THE ATMOSPHERE
REFLECTED
SOLAR RADIATION
SENSOR
TRANSMISSION
INTERACTION WITH
THE TARGET
RECEPTION AND
PROCESSING
BUILDINGS
ROAD
GRASS
WATER
FOREST
APPLICATIONS
INTERPRETATION
AND ANALYSIS

Light



Light travels in waves

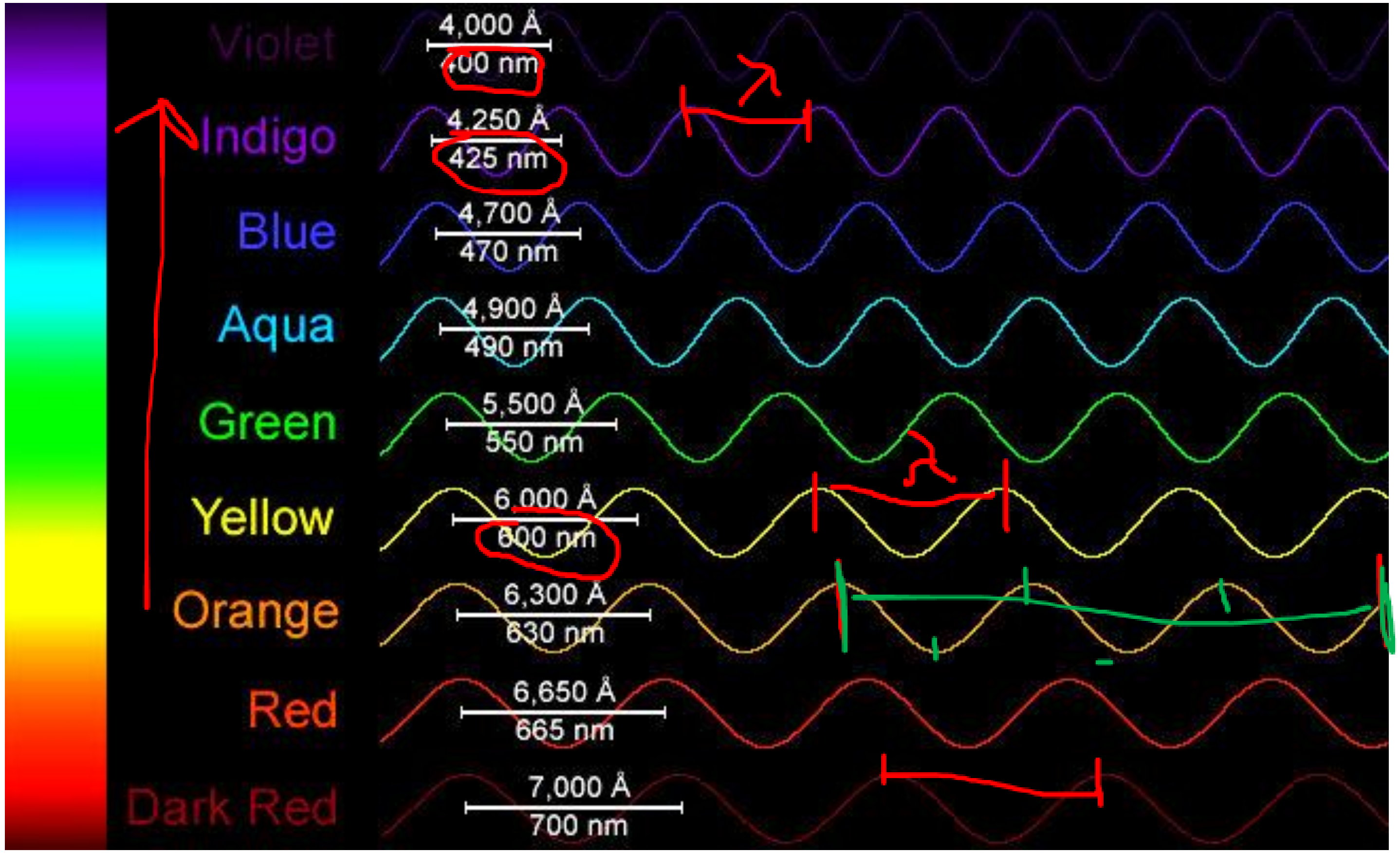


- Propagates as a bipolar wave.
- Has **electric** and **magnetic** properties.
- ~~Electrical properties are wave-based.~~
- Magnetic properties are particle-based (photons have no charge).
- ~~Moves at the speed of light.~~

Each color has a frequency and wavelength:

$c = \lambda v$

c = speed of light
λ = wavelength
v = frequency



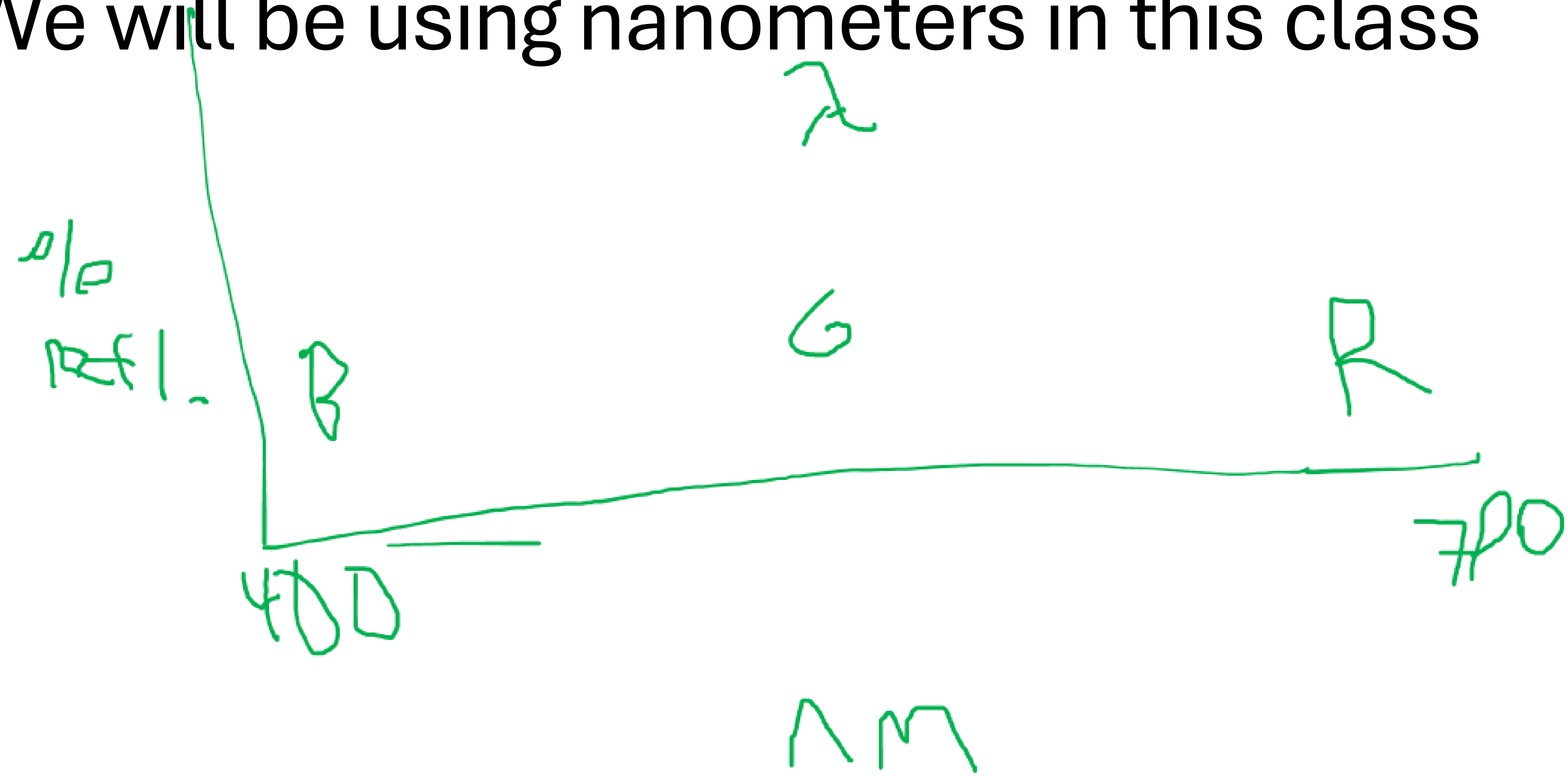
Wavelength: distance from peak to peak (λ).

~~Amplitude~~: height of the wave peak.

Frequency: the number of wave peaks passing a fixed point in space in unit time (v).

t 1 s

We will be using nanometers in this class

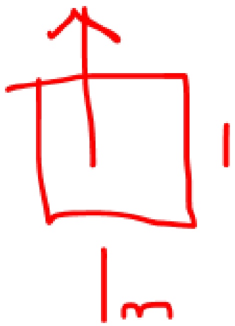


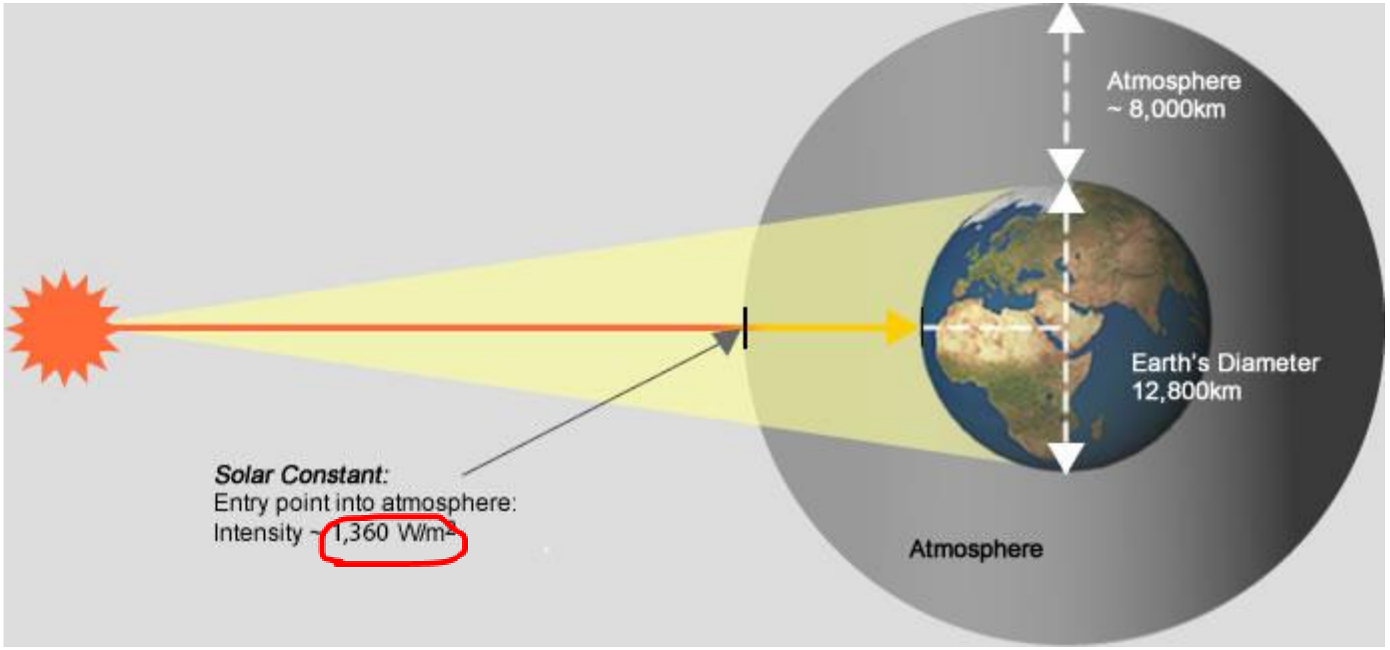
Split up into groups to fill in this table

Color	Wavelength range (nm)	Remote sensing applications
Ultraviolet	10nm – 400nm (shorter waveelgnths than we see)	Plant and Crop health, idefying bodily fluids, flower id, bugs, OIL SPILLS, atmosphere of our planet
Blue	400-500nm	Mapping snowmelt, Ocean, water state differentiation
Green	500-570nm	Bathymetry , forestry/crops/vegetation, pollution detection
Red	620-700nm	Chlroplhyll aborption, plants, plant stress, human-made features (roads), water quality
Far-red (red-edge)	700-800nm	Quality of plant growth,
Near infrared	800nm-1400nm	Vegetation growth and monitoring, soil properties, detecting water
Shortwave infrared	1400nm – 2500nm	Penetrate through atmosphere, water soil moisture, mineral composition, semiconductors, security and defense
Thermal infrared	~1,000,000nm	Temperature variation on the surface, forest firest, environmental polution
Microwave	1mm to meter	All weather conditions, day and night, subsurface data, large scale

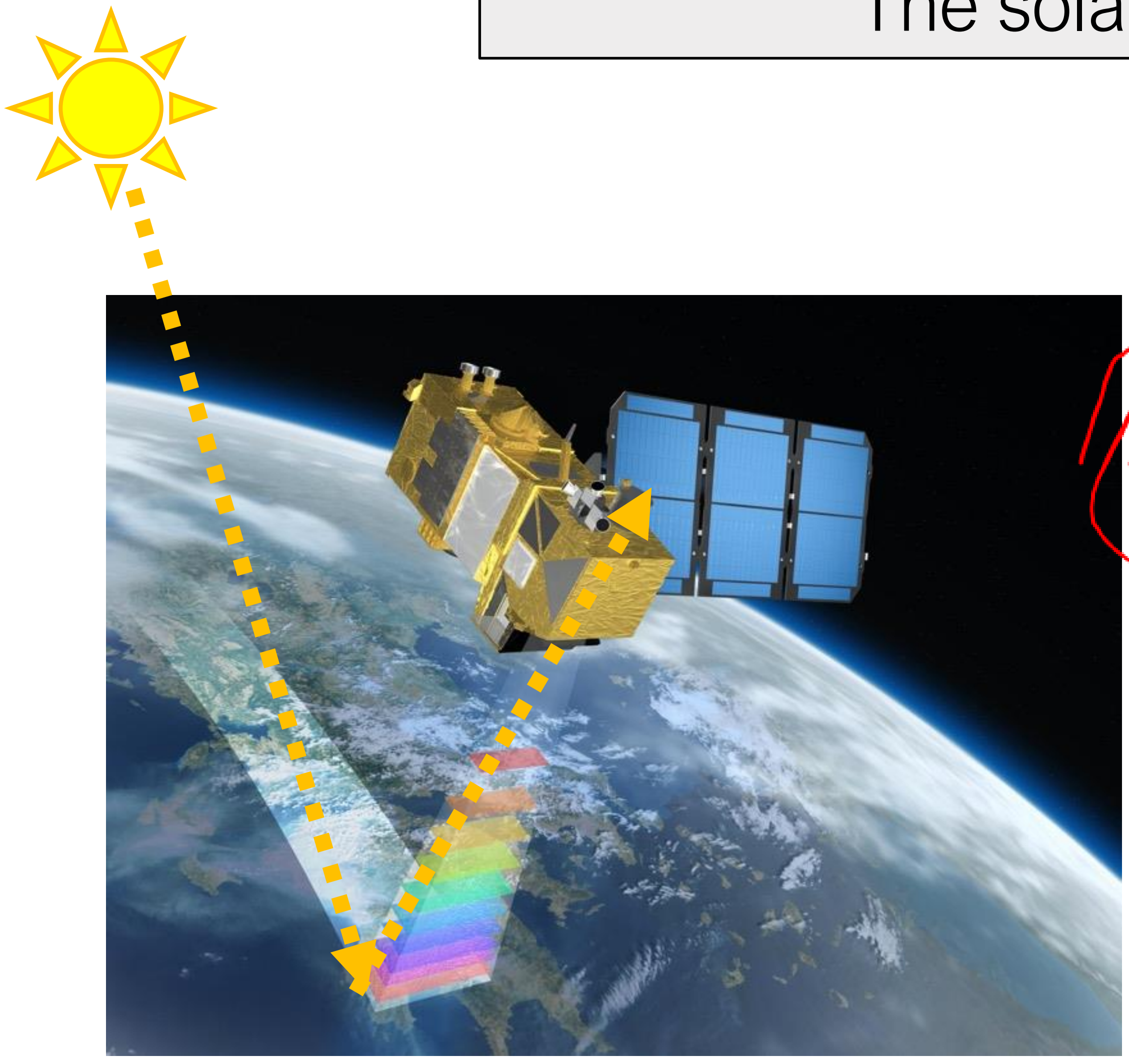
Solar Constant:

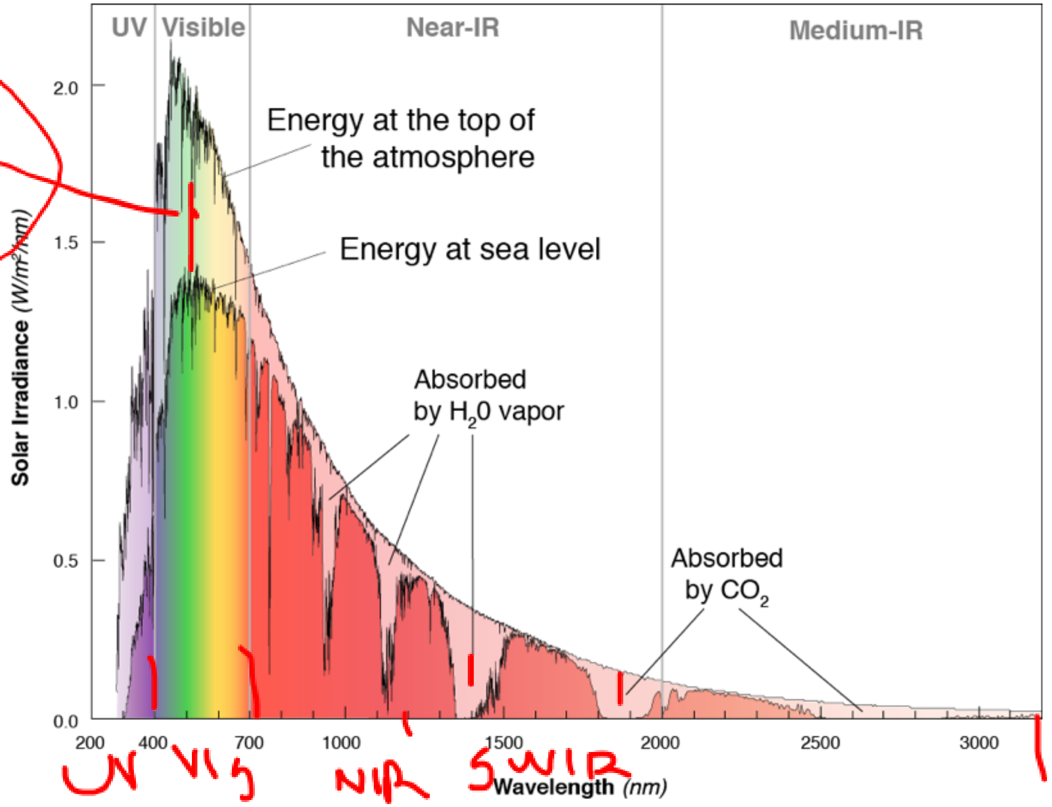
- 1.36 kWm^{-2} at top-of-atmosphere
- Varies ~6% in a year
- Clouds reduce surface incidence by 40-60%
- 550 – 1025 Wm^{-2} @ surface





The solar spectrum





Remote sensing is energy limited

- At long λ's there is little energy to sense.
- We have to 'image' large areas or for a longer time to have sufficient energy.
- This results in lower spatial resolution at $> \lambda$.